

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **end-to-end platform architectures** for robotics systems, integrating sensing, compute, and motion subsystems to create reusable building blocks.
- Developed and iterated on **rapid prototypes** encompassing electrical, mechanical, firmware, and software components to evaluate novel robotics concepts and de-risk system feasibility.
- Executed **characterization, benchmarking, and test automation** across hardware and software to generate data-driven insights for architecture choices and technology readiness assessments.
- Authored comprehensive **design docs, integration guides, and internal enablement materials** to effectively transfer system learnings and recommended patterns to product development teams.
- Wrote **Python scripts** for data analysis and automation, enhancing the efficiency of experimental data processing for robotics system characterization by an estimated **20%**.
- Contributed to the development of **Java Spring Boot REST APIs** for inter-service communication within enterprise systems, ensuring robust data exchange and integration.
- Managed and deployed containerised applications using **Docker** on **AWS EC2** instances, facilitating consistent testing and deployment environments for prototype systems.

SKILLS

Robotics & Systems – Robotic Systems Solutions (RSS), Systems Engineering, Platform Architecture, Rapid Prototyping, Hardware Bring-up, Firmware Development, Embedded Systems, Real-Time Development, Control Loops, System Integration, Connectivity, Telemetry, Edge Compute Integration, Safety, Motion Subsystems, Sensing Subsystems

Software Development – Python, Java, Spring Boot, REST APIs, Scripting, Data Analysis, Software Architecture, Debugging, Design Documentation, Integration Guides, Internal Enablement

Cloud & DevOps – AWS, EC2, S3, Lambda, CloudWatch, Docker, CI/CD, Jenkins, Git

Data & Analytics – Data Warehousing, Data Validation, SQL (PostgreSQL, MySQL, Oracle), Power BI, Tableau, DAX, Power Query

Methodologies & Tools – Agile, Scrum, Jira, Confluence, Linux, Shell Scripting

PROJECTS

Robotics Platform Integration Prototype

- Developed a Python-based data pipeline to ingest and process telemetry data from multiple simulated robotics subsystems, including sensor readings and motion commands.
- Integrated a basic control loop using Java to respond to simulated environmental inputs, demonstrating real-time responsiveness for a proof-of-concept system.
- Created initial design documentation outlining the proposed system architecture, highlighting potential integration challenges and data flow patterns for future development.

Automated Test Framework for Embedded Systems

- Engineered a Python framework to automate the execution of firmware tests on embedded hardware prototypes, reducing manual testing effort by approximately **40%**.
- Implemented data logging and analysis scripts to capture test results and performance metrics, facilitating faster debugging and iteration cycles for firmware engineers.
- Configured Jenkins pipelines to trigger automated test runs upon code commits, improving the efficiency of the continuous integration process for prototype development.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)